BARBARA FAGUNDES, Ph.D.
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EDUCATION

Purdue University Winter 2024
West Lafayette, IN
Ph.D., Engineering Education

Jackson State University Spring 2017
Jackson, MS
M.S., Computer Science

Universidade Sant'anna Summer 2010
São Paulo, Brazil
B.A., Early Childhood Education
Minors: K-5 education, policy, and school administration

TEACHING EXPERIENCE

University of Florida – Herbert Wertheim College of Engineering Gainesville, FL
Visiting Assistant Instructional Professor – Department of Engineering Education Fall 2026 – Present

- Teach undergraduate courses in Python Programming for Engineers and Engineering Design & Society
- Develop and adapt instructional materials, labs, assessments, and in-class activities for programming and engineering design
- Guide students through hands-on activities involving Arduino/Tinkercad, sensors and actuators, prototyping, flowcharts, introductory Onshape, and 3D printing
- Support students in programming, debugging, design decision-making, and project development through class activities and office hours
- Mentor student teams and collaborate with peer mentors and faculty to support active, project-based learning

Purdue University – School of Engineering Education West Lafayette, IN
Teaching Assistant, First-year Engineering Courses Fall 2019 – Fall 2021

- Supported instruction in two required First-Year Engineering courses serving ~240 students per semester
- Mentored student teams through semester-long engineering design projects
- Guided teams through problem scoping, stakeholder analysis, criteria and constraints development, and iterative design refinement
- Supported students in using Excel and MATLAB to evaluate design decisions
- Provided feedback on technical reports, visual communication, and oral presentations

- Led select class sessions as guest instructor
- Coordinated undergraduate TAs and graders, aligning grading practices and rubrics
- Held regular office hours supporting programming assignments, design challenges, and project development

Purdue University, School of Engineering Education West Lafayette, IN
Teaching Assistant, E-learning development (Graduate Course) Spring 2024 – Summer 2024

- Produced instructional videos and slides for the graduate course Teaching Engineering Online, recording and editing content prepared by the teaching team
- Assembled and organized course materials within the learning management system (LMS) for online delivery
- Coordinated course setup and logistics in collaboration with the teaching team
- Applied video production tools (Camtasia, Adobe Premiere Pro) to develop clear and engaging digital learning materials

Purdue University West Lafayette, IN
Teaching Assistant (Instructional Design) Summer 2021
CISTAR NSF Engineering Research Center

- Designed e-learning training modules for middle school teachers within NSF-funded engineering initiatives
- Collaborated with faculty and project partners to align instructional materials with learner needs and curriculum objectives
- Authored video scripts and produced instructional lessons and tutorials
- Utilized Camtasia and Adobe Premiere Pro to develop high-quality digital learning materials
- Completed instructional development within established timelines and project guidelines

Purdue University West Lafayette, IN
Teaching Assistant (Curriculum Development) Summer 2020
CISTAR NSF Engineering Research Center

- Collaborated with interdisciplinary research team members to develop a middle school chemical engineering curriculum
- Identified student learning needs through communication with faculty and iterative feedback cycles
- Established instructional goals and aligned activities with engineering learning objectives
- Participated in weekly progress meetings to refine curriculum design and implementation

Jackson State University Jackson, MS
Tutor of Computer Science Subjects Fall 2015 – Spring 2016
(Digital Logic, Physics, and Computer Science Courses)

- Provided academic support to undergraduate students in Digital Logic, Physics, and Computer Science courses
- Assisted students with programming assignments and lab activities
- Supported students in troubleshooting coding and logic-based problems

Pre-Kindergarten and Elementary School Teacher São Paulo, SP (Brazil)
Teacher in private and public schools Fall 2008 – Summer 2013
(Literacy, Mathematics, Geography, History, Science)

- Designed and delivered interdisciplinary lessons in literacy, mathematics, science, geography, and

history

- Implemented project-based learning activities to support early problem-solving and collaboration skills
- Differentiated instruction for students with different learning needs and backgrounds.
- Conducted monthly progress assessments and prepared detailed reports for parents and school administration
- Led parent meetings and participated in faculty coordination meetings

RESEARCH EXPERIENCE

Purdue University West Lafayette, IN
Postdoctoral Researcher Fall 2024 – Summer 2026
Center for Instructional Excellence (CIE)

- Led qualitative research on undergraduate students' use of generative AI in computer science and engineering courses
- Designed and facilitated focus groups and interviews; analyzed qualitative and survey data
- Co-authored manuscripts and presentations translating findings into evidence-based teaching recommendations
- Contributed to projects examining learner motivation and autonomy using Self-Determination Theory
- Co-led the Student Pedagogy Advocates (SPA) program, mentoring undergraduate partners in collaborative teaching and learning initiatives
- Co-mentored undergraduate partners in collaborative projects supporting faculty teaching innovation and student learning

Purdue University West Lafayette, IN
Graduate Research Assistant at the NSF RECT project Spring 2022 – Fall 2023
(Rethinking Circle Time: Integrating Computational Thinking into K-2 Literacy)

- Developed programming lessons integrating computational thinking into K–2 literacy curriculum
- Created instructional video tutorials including scripting, filming, and editing
- Conducted classroom-based video data collection and qualitative analysis
- Performed literature reviews in K–12 STEM and computational thinking education
- Co-authored peer-reviewed journal and conference papers
- Designed and administered Qualtrics surveys to assess teachers' computational thinking implementation
- Mentored junior PhD students and supported collaborative research activities

Purdue University West Lafayette, IN
Graduate Research Assistant Fall 2018 – Fall 2019
(Epistemic Research Laboratory)

- Coordinated classroom-based video data collection in K–12 engineering settings
- Managed research logistics to ensure timely and compliant data collection
- Supported implementation of engineering design curriculum in K–12 classrooms
- Co-authored peer-reviewed conference paper on engineering design decision processes

Purdue University West Lafayette, IN
Data Research Manager Fall 2017 – Fall 2018
(Research Institute for Pre-College Engineering - INSPIRE)

- Managed research data and classroom-based video collection for grades 6–8
- Coordinated logistics for large-scale pre-college engineering research initiatives
- Conducted computational thinking activities for K–12 students
- Collaborated with faculty and graduate researchers to streamline research

JOURNAL AND PEER-REVIEWED CONFERENCE PUBLICATIONS

Fagundes, B., & Moore, T. J. (2025, June). *Comparing computational thinking learning and engagement in first-grade boys and girls: A study of algorithm design and debugging (WIP).* In *Proceedings of the ASEE Annual Conference & Exposition*, Pre-College Engineering Education Division, Montréal, Canada. *(This paper draws from Dr. Fagundes's doctoral dissertation research on engagement and computational thinking in early childhood education.)*

Fagundes, B., Guberman, D., Smart, K., & Holder, K. (2025, June). *Student Pedagogy Advocates: Enhancing teaching and learning through student–faculty partnerships (WIP).* In *Proceedings of the ASEE Annual Conference & Exposition*, Faculty Development Division, Montréal, Canada. https://doi.org/10.18260/1-2--55986 *(This work builds on Dr. Fagundes's postdoctoral research on faculty development and autonomy-supportive teaching practices.)*

Tank, K. M., Ottenbreit-Leftwich, A., Moore, T. J., Yang, S., Wafula, Z., Kim, J., **Fagundes, B.,** & Chu, L. (2024). Investigating Sequencing as a Means to Computational Thinking in Young Children. *International Journal of Computer Science Education in Schools*, *6*(3), 67–77. https://doi.org/10.21585/ijcses.v6i3.192

Fagundes, B., Bhide, N., Moore, T. J., & Tank, K. M. (2021, July*), Computational Thinking in First Grade Students Using a Computational Device (WIP)* Paper presented at 2021 ASEE Virtual Annual Conference Content Access, Virtual Conference. 10.18260/1-2--36825

Fagundes, B., Ehsan, H., Moore, T. J., Tank, K. M., & Cardella, M. E. (2020, June), *First-graders' Computational Thinking in Informal Learning Settings (WIP)*. Paper presented at 2020 ASEE Virtual Annual Conference Content Access, Virtual Online. 10.18260/1-2--35541.

Johnston, A. C., Lopez-Parra, R. D., Tank, K. M., Moore, T. J., & **Fagundes, B.** (2019, June), *Design Decision Processes of First Grade Students during an Engineering Design-based STEM Unit (Fundamental)* Paper presented at 2019 ASEE Annual Conference & Exposition, Tampa, Florida. 10.18260/1-2--32599

Yeter, I. H., Rynearson, A. M., Ehsan, H., Rehmat, A. P., Dasgupta, A., **Fagundes, B.,** Menekse, M., & Cardella, M. E. (2019, June), *Design and Implementation of Data Collection in a Large-Scale, Multi-Year Pre-College Engineering Study: A Retrospective.* Paper presented at 2019 ASEE Annual Conference & Exposition, Tampa, Florida. 10.18260/1-2--32596

PREPRINTS AND MANUSCRIPTS IN PREPARATION

Dickey, E., Bejarano, A., Kuperus, R., & **Fagundes, B.** (2025). *Evaluating the AI-Lab intervention: Impact on student perception and use of generative AI in early undergraduate computer science courses. arXiv preprint* arXiv:2505.00100.

Fagundes, B., Harris-Thomas, B., Bonem, E., Guberman, D., & Levesque-Bristol, C. (2026). *Supporting student autonomy in higher education STEM classrooms: Faculty practices, barriers, and evolving perceptions.* [Manuscript in preparation]. Target journal: *International Journal of STEM Education.*

PRESENTATIONS, WORKSHOPS, AND POSTERS

Moore, T., Tank, K., & **Fagundes, B.** (2023). *Task Interviews as a Research Tool: Cases from K–2 Computational Thinking.* Workshop presented at the Pre-College Engineering Education Division (PCEE) at the 2023 ASEE Annual Conference & Exposition, Baltimore, MD.

Ottenbreit-Leftwich, A., Moore, T. J., Tank, K. M., Kim, J., **Fagundes, B.,** Chu, L., & Wafula, Z. (2023). *Rethinking Circle Time: Development of K-2 CT Literacy Integrated Curriculum* [Poster presentation]. SIGCSE 2023: 54th ACM Technical Symposium on Computer Science Education, Toronto, Ontario, Canada. https://doi.org/10.1145/3545947.3576338. (pp. 1394-1394).

Fagundes, B., Bhide, N., Moore, T. J., Drummond Oaks, M., & Godwin, A. (2022, June). *Online professional development aid for teaching an engineering design-based curriculum in 8th grade (Resource Exchange)*. Work showcased at the Exhibit Hall in 2022 ASEE Annual Conference & Exposition, Minneapolis, MN.

Lopez-Parra, R. D., **Fagundes, B.,** Wallace, D., Bhide, N., Moore, T. J., Drummond Oaks, M., & Godwin, A. (2022, June). *Engineering design-based curriculum for teaching 8th grade students renewable energy (Resource Exchange)*. Work showcased at the Exhibit Hall in 2022 ASEE Annual Conference & Exposition, Minneapolis, MN.

Fagundes, B., & Bhide, N. (2021, July). *Computational Thinking in First-Grade Students Using a Computational Device.* Virtual Poster presentation at 2021 ASEE Annual Conference & Exposition. Online.

Fagundes, B., & Cardella, M. (2018, December). *Exploring Engineering Behaviors of Young Children Interacting with a Parent.* ENE Explorer Poster Session Presentation at Engineering Education Researcher Seminar. Purdue University in West Lafayette, IN.

HONORS AND AWARDS

● **Outstanding Service Scholarship Award,** Purdue University College of Engineering, 2024

Recognized by the College of Engineering for outstanding service and contributions to the Purdue community.

● **Purdue Society of Women Engineers (SWE)** – WE22 National Conference

Selected and sponsored by Purdue SWE to represent the university at the 2022 Women in Engineering National Conference.

● **Grace Hopper Celebration (GHC)** Student Scholar, Orlando, FL, 2019

Selected as a GHC Scholar to attend the Grace Hopper Celebration, the world's largest conference for women in computing.

● **Diversity Scholarship**, Jackson State University, 2016

Awarded a merit-based scholarship supporting international students from underrepresented backgrounds.

● **Presidential Scholarship for Minority Students (PROUNI – University for All Program),** Universidade Sant'Anna, São Paulo, Brazil, 2007

Awarded a full-tuition federal scholarship for undergraduate studies through Brazil's national higher education inclusion program, based on academic excellence and performance on the National High School Exam (ENEM).

ACADEMIC AND PROFESSIONAL SERVICE

● **Reviewer, Journal of Pre-College Engineering Education Research (J-PEER)** Fall 2025–Present (peer-reviewed journal in engineering education)

● **Brazilian Student Association (BRASA Global)** – Student Brand Ambassador

Summer 2022 – Summer 2023

Volunteer speaker giving virtual presentations at Brazilian universities to promote international scholarships, mentorship programs, and educational opportunities for low-income students. Collaborated with other BRASA ambassadors worldwide to organize crowdfunding initiatives supporting scholarship programs for students with limited financial resources.

● **Women in Engineering Program (WIEP)** – Pre-College Engineering Outreach Team Member, Purdue University - *Summer 2022 – Summer 2023*

Mentored pre-college girls in engineering through on- and off-campus activities. Collaborated with a team of women engineering students to promote early engagement and interest in STEM fields.

● **Purdue Graduate Student Government (PGSG)** – Senior Senator

Fall 2022 – Spring 2023

Represented the Engineering Education Graduate Student Association (ENEGSA) in PGSG Senate meetings. Contributed to discussions, resolutions, and policy proposals affecting graduate and professional students. Reported outcomes to ENEGSA members to support transparency and shared governance.

● **Grant Review and Allocation Committee (GRAC)** – Reviewer, PGSG

Fall 2022

Reviewed and evaluated Purdue students' applications for professional development and community-based grants. Provided feedback and funding recommendations based on merit and alignment with program goals.

● **American Society for Engineering Education (ASEE)** – Reviewer

Fall 2018 – Present

Evaluate annual ASEE conference submissions for rigor, clarity, and contribution to the engineering education community. Provide detailed feedback to authors and editorial recommendations on acceptance and revision.

● **Engineering Education Graduate Student Association (ENEGSA)** – Member, ESL & Multicultural Committee, Purdue University, *2020*

Co-created and hosted *"We, the Aliens"* podcast, discussing the experiences of international students in U.S. higher education. Organized workshops and collaborated with Purdue's Latino Cultural Center to foster intercultural dialogue.

● **Summer Undergraduate Research Fellowship (SURF) e-Symposium** – Judge

2020

Served as a judge for undergraduate research poster presentations in collaboration with the Engineering Undergraduate Research Office (EURO). Evaluated and provided feedback on research quality and presentation effectiveness during the first fully virtual SURF Symposium.

● **School of Engineering Education (ENE), Purdue University** – Academic Mentor

2020

Mentored Ph.D. students in the School of Engineering Education, offering one-on-one guidance on research design, coursework, and teaching practices during the COVID-19 transition to online learning.

PROFESSIONAL MEMBERSHIPS

● Member, American Society for Engineering Education (ASEE)
● Society of Women Engineers (SWE)

LANGUAGES AND SOFTWARE

Programming Languages: Python, MATLAB, JavaScript, Java, C++, PHP, SQL
Engineering and Design Tools: Arduino, Tinkercad, introductory Onshape
Digital and Web Tools: HTML, CSS, WordPress, MS Office, LMS tools, Camtasia, Articulate 360, Adobe Premiere Pro, Adobe InDesign